

Understanding Dataset Difficulty with V-Usable Information

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Why We Need Better Difficulty Measures

Everyday NLI scenario (SNLI):

Premise	A man is playing a guitar on stage.
Hypothesis	A person is performing music.
Gold	Entailment
Model prediction	Neutral (confused by wording)

- Accuracy alone says “wrong,” but not why this instance is hard.
- We need a per-instance, model-aware signal of usable information.

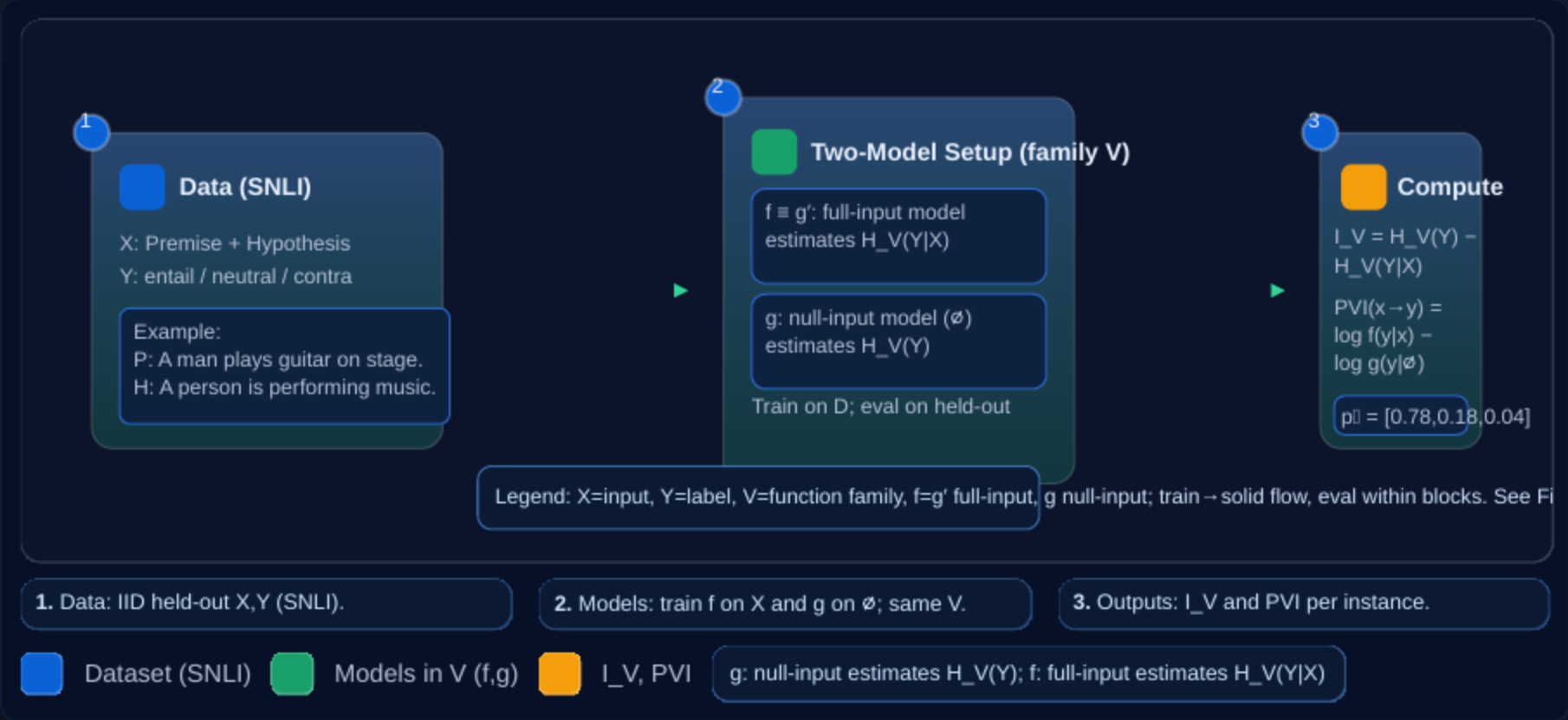
Abstract & Motivation

- Define dataset difficulty for a model family V as lack of V -usable information.
- Introduce pointwise V -information (PVI) to score instance-level difficulty.
- Compare models on a dataset and datasets for a given model; analyze slices.
- Use input transformations to interpret attribute utility and uncover artefacts.

V-Usable Information: Concept & Setup

- Predictive V -information $I_V(X \rightarrow Y)$: info usable by model family V (Xu et al., 2019).
- $I_V(X \rightarrow Y) = H_V(Y) - H_V(Y|X)$; higher $\rightarrow$ easier dataset for V .
- Estimate via two models in V : with input X and with null input $\emptyset$ on IID holdout.
- Properties: non-negativity, independence, monotonicity ($U \subseteq V \Rightarrow I_U \leq I_V$).

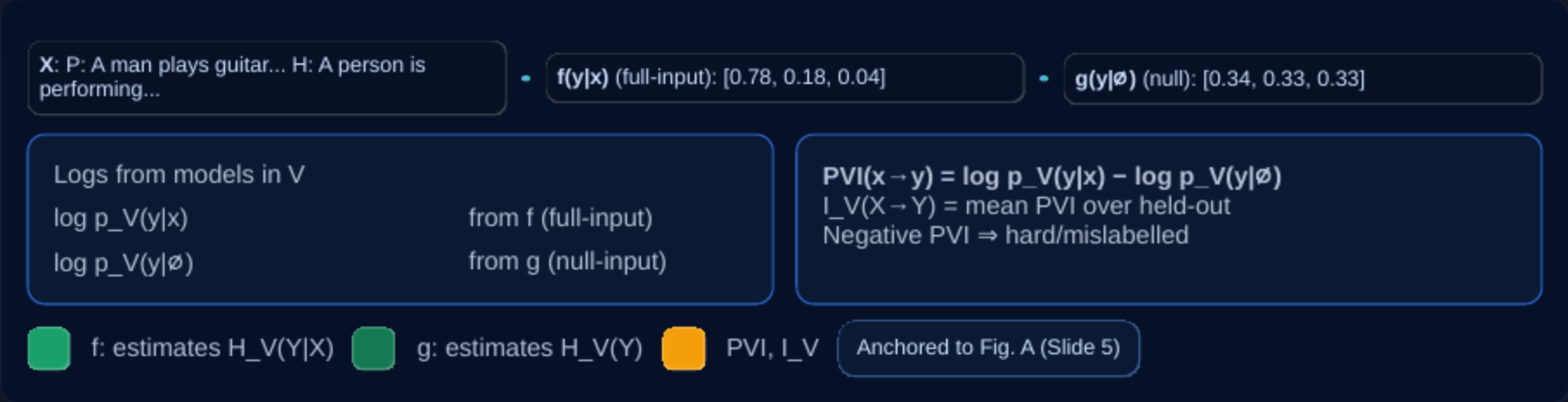
Fig. A — End-to-End: Data → Two-Model Setup → Compute V/PVI



High-contrast pipeline (Fig. A)

Slide 5

Fig. B — Zoom: How PVI Is Computed (see Fig. A)



PVI node detail (Fig. B)

Slide 6

Pointwise V-Information (PVI)

- $PVI(x \rightarrow y) = \log p_V(y|x) - \log p_V(y|\emptyset)$; average equals $I_V(X \rightarrow Y)$ — see Fig. B.
- Higher $\rightarrow$ easier for V; can be negative (e.g., misleading cues, mislabels).
- On SNLI, correct items have much higher PVI; threshold ≈ 0.5 bits.
- Stable across models/seeds on SNLI (high correlations).

See Fig. A (Slide 5) for pipeline $\rightarrow$ PVI, Fig. B (Slide 6) for zoom.

SNLI: What Attributes Carry Usable Information?

- Token identity (order-agnostic via shuffling) carries most usable info.
- Premise-only is near useless; hypothesis-only retains substantial info.
- Overlap tokens are especially helpful for entailment vs other classes.

Mean PVI by class (original)	Entailment 1.188 Neutral 1.064 Contradiction 1.309
Shuffled	Similar magnitudes $\rightarrow$ order matters less than identity
Hypothesis / Premise	Hypothesis-only informative; premise-only ≈ 0

SNLI Slices via Mean PVI

- Use mean PVI over slices for relative difficulty (not V-info of slice).
- By overlap length: entailment with no overlap → hardest; contradiction with no overlap → easiest.
- Cartography regions align with distinct PVI ranges.

Conclusions

- V-information quantifies dataset difficulty for a model family.
- PVI provides stable, instance-level difficulty estimates on SNLI.
- Attribute probes reveal artefacts; guide principled dataset curation.